CS & DA





Probability and Statistics

Permutations and Combinations

Lecture No. 02



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Recap of previous lecture



Topic Counting Principle

Topics to be Covered



Topic

BASICS of P & C (Part 2)

Perm & Combination & (Counting Principle)

Fundamental Principle of Addition → if we have to perform only one of the job at a time out of m jobs then use this principle. Keywords: "Either or / only one / Anyone"

Fundamental Principle of Multiplication → if we have to perform all the jobs at a time (out of m jobs) then use this Principle

Keywords: "AND / Both / All"

Q: There are 8 chairs then in how many ways 5 persons can be seated?

Sol: Total Seating Arrangements = $\underset{P_1}{8 \text{ ways}} \times \underset{P_2}{7 \text{ ways}} \times \underset{P_3}{6 \text{ ways}} \times \underset{P_4}{5 \text{ ways}} \times \underset{P_5}{4 \text{ ways}}$

(RNA)

$= \underbrace{8 \times 7 \times 6 \times 5 \times 4}_{\text{M. Rule}} = {}^{8}C_5 \times 5! = \underbrace{{}^{8}P_5}_{\text{P. Rule}}$

(ii) If No. of chairs = (6) & No. of persons = (8) then T.S.A = ? $= 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$

(RNA) $= {}^{8}P_8 = 8!$

(iii) If No. of Chairs = 5 & No. of persons = 8 then T.S.A = ? = Senseless Question

Conclusion:- Let n = No. of places, r = No. of persons

① if $n > r$ & RNA then Multi Rule = Perm Rule

② if $n = r$ & RNA then Multi Rule = Perm-Rule = Factorial Rule

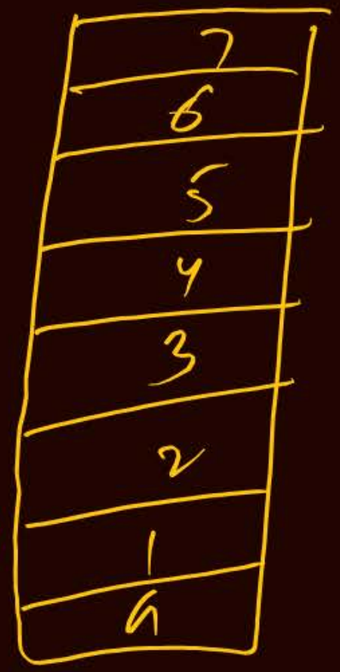
③ if RA then only use Multi Rule.

is the concept of ${}^nC_r$, ${}^nP_r$ & $r!$ exist only when R.N.A

(HW8) 5 persons entered in a lift at Ground floor in 8 floor house.
Then in how many ways they can leave the lift

① At any floor = ? (RA)
$= \underset{P_1}{7} \times \underset{P_2}{7} \times \underset{P_3}{7} \times \underset{P_4}{7} \times \underset{P_5}{7} = 7^5$ ways

② At different floors = ? (RNA)
$= \underset{P_1}{7} \times \underset{P_2}{6} \times \underset{P_3}{5} \times \underset{P_4}{4} \times \underset{P_5}{3} = {}^{7}P_5$ ways



Combination → (when ever Counting is Based on only selection then we will use Combination Rule)

$$^nC_r = \frac{n!}{r!(n-r)!}$$ = selection of r things from n things

Note: we know that $^nC_r = {}^nC_{n-r}$

for eg $^{22}C_{18} = ? = {}^{22}C_{22-18}$

$= {}^{22}C_4 = \frac{22\times21\times20\times19}{4\times3\times2\times1}$

shortcut: $^{11}C_3 = \frac{11\times10\times9}{3\times2\times1}$

$^{22}C_4 = \frac{22\times21\times20\times19}{4\times3\times2\times}$

eg $^7C_2 = \frac{7\times6}{2\times1}$, $^6C_1 = 6$, $^7C_1 = 7$

eg $^5C_1 = \frac{5}{1} = 5$, $^6C_0 = 1$, $^7C_0 = 1$

$^nC_0 = 1$, $^nC_1 = n$, $^nC_2 = \frac{n(n-1)}{2\times1}$, $^nC_n = {}^nC_0 = 1$, $^nC_0 = \frac{n!}{0!(n-0)!} = \frac{n!}{n!} = 1$

Note: $0! = ? = 1$ $= \frac{1}{1}$

$1! = 1 = 1 = \frac{2}{2}$

$2! = 1 \times 2 = 2 = \frac{6}{3}$

$3! = 1 \times 2 \times 3 = 6 = \frac{24}{4}$

$4! = 1 \times 2 \times 3 \times 4 = 24 = \frac{120}{5}$

$5! = 1 \times 2 \times 3 \times 4 \times 5 = 120$

$\vdots$

Q $^{15}C_{12} = ?$

M-I $^{15}C_{12} = \frac{15 \times 14 \times 13 \times 12 \times 11 \times 10 \times \ldots \text{Tired}}{12 \times 11 \times 10 \times 9 \times 8 \times \ldots \text{again Tired}}$

M-II $^{15}C_{12} = ? = {}^{15}C_{15-12} = {}^{15}C_3$

$= \frac{15 \times 14 \times 13}{3 \times 2 \times 1} = \text{Ans}$

$^{31}C_{19} = ? = {}^{31}C_{12} = \text{Again lengthy}$

Permutation → when Counting is Based on Selection & Arrangement Both

(then we will use Perm. Rule) → **Perm = Selection & Arrangement**

$$^nP_r = \frac{n!}{(n-r)!} = {}^nC_r \times r!$$

^nC_r: Selection, $r!$: Arrangement.

eg $^{11}P_3 = 11\times10\times9$, $^{22}P_4 = 22\times21\times20\times19$

$^{11}C_3 = \frac{11\times10\times9}{3\times2\times1}$

$^nP_0 = {}^nC_0 = 1$, $^nP_1 = n = {}^nC_1$

$^7P_2 = 7\times6$, $^7P_1 = 7$

$^7C_2 = \frac{7\times6}{2\times1}$, $^7C_1 = \frac{7}{1} = 7$

$^7C_0 = 1 = {}^7C_0$

$^nP_n = n!$, $^nC_n = 1$, $^nP_2 = n(n-1)$, $^nC_2 = \frac{n(n-1)}{2\times1}$

Ex of Combinations :-

eg formation of Committee, Number of Handshakes

eg " " team ; No. of straight lines, No. of Δ's etc

Ex of Permutations -

eg formation of words

eg " " Numbers

eg " " Signals

eg seating Arrangements

eg formation of Photographs

"Whenever in a Question, there is a feeling of Interchanging things then we will use Perm."

Q. In a party there are 5 persons then how many different Handshakes are possible in that party?

(M-I) Various Handshakes: $(P_1P_2), (P_1P_3), (P_1P_4), (P_1P_5), (P_1P_1)$

(RNA) $(P_2P_1), (P_2P_2), (P_2P_3), (P_2P_4), (P_2P_5)$

$(P_3P_1), (P_3P_2), (P_3P_3), (P_3P_4), (P_3P_5)$

(P_4P_5) ∴ Total H-Shakes = 10

(M-II) ∵ Counting of H-Shakes is Based on only Selection ∴ Total H-Shakes $= {}^5C_2$

$= \frac{5 \times 4}{2 \times 1} = 10$

Q. Salads are made from one or more vegetables then how many different salads can be made using onion, Tomatto, Reddish, Cucumber, Beetroot?

Sol: He or she can make salad either by

selecting 1V or 2V or 3V or 4V or 5V

$= {}^5C_1 + {}^5C_2 + {}^5C_3 + {}^5C_4 + {}^5C_5$

$= 5 + 10 + 10 + 5 + 1 = 31$ salads

(ii) How many different salads can be made using exactly 3 vegies at a time = ? $= {}^5C_3 = 10$

(iii) " " " " " " " " " at least 3 " " " = ? $= {}^5C_3 + {}^5C_4 + {}^5C_5$

(iv) " " " " " " " " " at most 3 " " " = ? $= {}^5C_1 + {}^5C_2 + {}^5C_3 = 5 + 10 + 10 = 25$

$= 10 + 5 + 1 = 16$

Part (II) : O, T, R, C, B , No. of ways to deal with each Vegie = 2 ways (either S or R)

" " " " " all Vegies = $\underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2}$
O T R C B

$= 2^5 = 32$ Salads

Here we have also included Imaginary Salad
So Req. Ans = 32 − 1 = 31 Salads

(SSSSS) ⟶ ${}^5C_5 = 1$ Salad
(SSSSR) ⟶ ${}^5C_4 = 5$ Salads
(SSSRR) ⟶ ${}^5C_3 = 10$ Salads
(SSRRR) ⟶ ${}^5C_2 = 10$ Salads
(SRRRR) ⟶ ${}^5C_1 = 5$ Salads
(RRRRR) ⟶ ${}^5C_0 = 1$ Salad

Imaginary Salad

32 Salads

P.Q A Person has 6 friends. In how many ways he or she can organise B'day Party?

(a) $6!$ $f_1, f_2, f_3, f_4, f_5, f_6$ Most

(b) 6 Person can organise B'day Party either by Inviting

$= (0f \text{ or } 1f \text{ or } 2f \text{ or } 3f \text{ or } 4f \text{ or } 5f \text{ or } 6f)$

(c) $^7P_6$ $= X$ $= {}^6C_1 + {}^6C_2 + {}^6C_3 + {}^6C_4 + {}^6C_5 + {}^6C_6$

(d) ✓ 63 $= 6 + 15 + 20 + 15 + 6 + 1 = 63$

(M-II) Total ways of Organising party $= ? = \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} \times \underline{2} = 2^6 = 64$ So Ans $= 63$

$f_1\ f_2\ f_3\ f_4\ f_5\ f_6$

(SSSSSS), (SSSSSR), (SSSSRR) (SSSRRR), (SSRRRR), (SRRRRR), (RRRRRR)

${}^6C_6 = 1$ ${}^6C_5 = 6$ ${}^6C_4 = 15$ ${}^6C_3 = 20$ ${}^6C_2 = 15$ ${}^6C_1 = 6$ ${}^6C_0 = 1$

Q. There are 5 flags of different colours then how many different signals can be made using

① 3 flags at a time = ? $= \underbrace{5 \text{ ways}}_{P_1} \times \underbrace{4 \text{ ways}}_{P_2} \times \underbrace{3 \text{ ways}}_{P_3}$ (M-I)

$= {}^5C_3 \times 3! = {}^5P_3$ (M-II, M-III)



↓ Selection of 3 flags

↓ Arrangement of 3 selected flags

② Any Number of flags at a time = ?

We can make signal either By selecting = 1f or 2f or 3f or 4f or 5f

(M-II) $= ({}^5C_1 \times 1!) + ({}^5C_2 \times 2!) + ({}^5C_3 \times 3!) + ({}^5C_4 \times 4!) + ({}^5C_5 \times 5!)$

(M-III) $= {}^5P_1 + {}^5P_2 + {}^5P_3 + {}^5P_4 + {}^5P_5 = 5 + 20 + 60 + 120 + 120 = 325$

Q. out of 6 Gents & 4 Ladies, a Committee of 5 persons is to be formed then find the Number of Committees if

① there is No Restriction =? (RNA) $= {}^{10}C_5 = \frac{10\times9\times8\times7\times6}{5\times4\times3\times2\times1} = 252$ Committees (Max Ans).

② there are at Most 2L in a Committee =? $= (0L\&5G) \text{ or } (1L\&4G) \text{ or } (2L\&3G)$

$= ({}^4C_0 \times {}^6C_5) + ({}^4C_1 \times {}^6C_4) + ({}^4C_2 \times {}^6C_3) = 186$

③ there are At least 2L in a Committee =? $= (2L\&3G) \text{ or } (3L\&2G) \text{ or } (4L\&1G) \text{ or } (5L\&0G)$

$= ({}^4C_2 \times {}^6C_3) + ({}^4C_3 \times {}^6C_2) + ({}^4C_4 \times {}^6C_1) = 186$

④ Gents are in Majority =? $= (5G\&0L) \text{ or } (4G\&1L) \text{ or } (3G\&2L)$

$=$ Same as Part 2 $= 186$

(5) Ladies are in Majority = ? = (4L & 1G) or (3L & 2G)

$= ({}^4C_4 \times {}^6C_1) + ({}^4C_3 \times {}^6C_2) = 66$

(6) At least one Lady must be there = ?

(M-I) Req Ans = (1L & 4G) or (2L & 3G) or (3L & 2G) or (4L & 1G)

$= ({}^4C_1 \times {}^6C_4) + ({}^4C_2 \times {}^6C_3) + ({}^4C_3 \times {}^6C_2) + ({}^4C_4 \times {}^6C_1) = 246$

(M-II) At least one Lady = ? = Total − No Lady

= Total − All Gents

$= {}^{10}C_5 - {}^6C_5 = 252 - 6 = 246$

Note:

At least one = Total − None

Analysis:- 6G & 4L & Committee of 5 Persons is to be formed

Case I (0L & 5G) or Case II (1L & 4G) or Case III (2L & 3G) or Case IV (3L & 2G) or Case V (4L & 1G) = Total committees

$({}^{4}C_0 \times {}^{6}C_5) + ({}^{4}C_1 \times {}^{6}C_4) + ({}^{4}C_2 \times {}^{6}C_3) + ({}^{4}C_3 \times {}^{6}C_2) + ({}^{4}C_4 \times {}^{6}C_1) = {}^{10}C_5$

$(1 \times 6) + (4 \times 15) + (6 \times 20) + (4 \times 15) + (1 \times 6) = 252$

$6 + 60 + 120 + 60 + 6 = 252$

At Most 2L

At least 3L

At least 2L

At least one L

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THANK - YOU